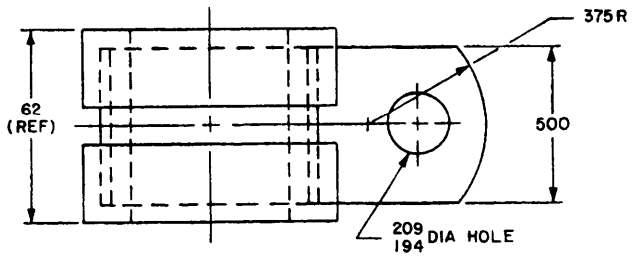
MIL-C-85052/3B
30 MAY 1984
SUPERSEDING
MIL-C-85052/3
15 SEPTEMBER 1980

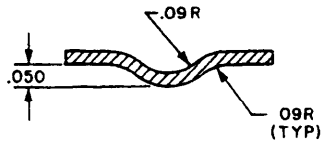
MILITARY SPECIFICATION SHEET

CLAMP, LOOP, TUBE - 17-7 PH CRES, 500°F

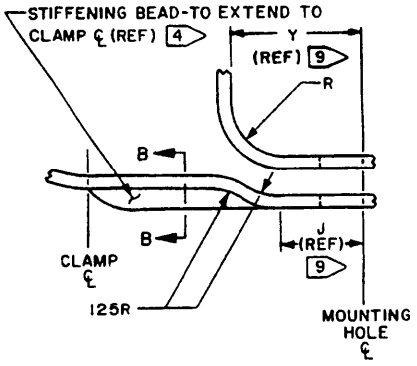
THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

THE COMPLETE REQUIREMENTS FOR ACQUIRING THE CLAMPS DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE IN EFFECT OF SPECIFICATION MIL-C-85052

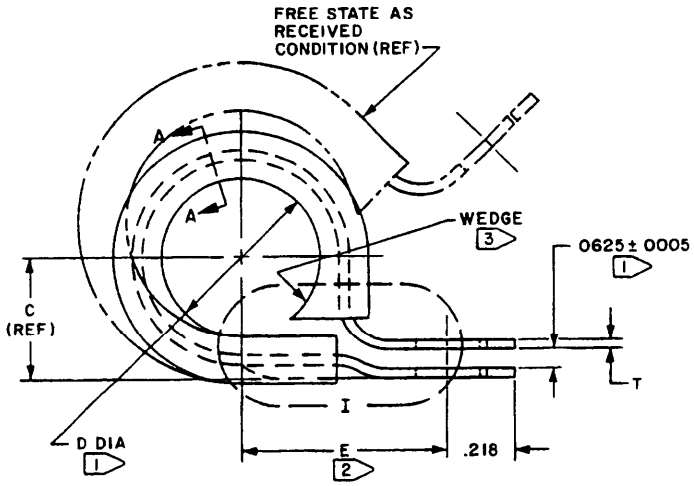




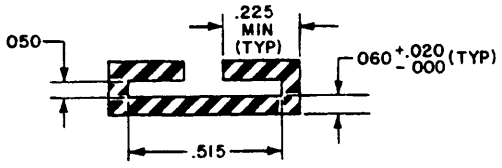
SECTION B-B



DETAIL I
(CUSHION REMOVED)



TOLERANCES XXX ± 010, .XX ± .03 UNLESS OTHERWISE SPECIFIED



SECTION A-A
(CUSHION ONLY)

(B) DENOTES CHANGES

FSC 5340

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TABLE I

DASH NUMBER	C (REF)	D DIA (1)	(B) E ±.032 (2)	J (REF)	R ±.010	T	Y (REF)
2	192	125	.468	--			
3	224	188	499	--			
4	255	.250	530			020	
5	286	312	561		090	±.002	325
6	.318	375	592	235			
7	349	438	624				
8	380	500	655				
9	423	.562	741				
10	454	625	772			032	
11	486	688	804	256	125	±.002	368
12	517	.750	835				
13	548	812	866				

DASH NUMBER	C (REF)	D DIA (1)	(B) E ±.032 (2)	J (REF)	R ±.010	T	Y (REF)
14	580	875	898				
15	611	938	929	256		032	368
16	642	1 000	960			±.002	
17	681	1 062	1 001				
18	712	1 125	1 032				
19	744	1 188	1 064		125		
20	775	1 250	1 095			040	
21	806	1 312	1 126	262		±.003	370
22	838	1 375	1 158				
23	869	1 438	1 189				
24	900	1 500	1 220				

NOTES

(1) DIAMETER "D" IS THE NOMINAL DIAMETER FOR WHICH A CLAMP SIZE IS INTENDED FOR USE DIAMETER "D" SHALL BE VERIFIED BY THE "DIAMETRAL RETENTION" TEST OF THE GENERAL SPECIFICATION

(2) DIMENSION "E" SHALL BE MEASURED WITH THE CLAMP INSTALLED ON A MANDREL OF "D" DIAMETER ±.001 AND AN .0625 ±.0005 SPACER BETWEEN THE UPPER AND LOWER FOOT AS SHOWN

(B) (3) WEDGE SHALL BE REQUIRED ON SIZE -6 AND LARGER THE WEDGE SHALL BE MOLDED AS AN INTEGRAL PART OF THE CUSHION AND CONTOURED TO FIT "D" DIAMETER.

(B) (4) STIFFENING BEAD SHALL BE REQUIRED ON SIZE -4 AND LARGER ALL RADII OF THE STIFFENING BEAD SHALL BE SMOOTH AND BLENDED. NO SHARP TOOL MARKS ARE ALLOWED

(B) 5 UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.

6 METAL BANDS SHALL HAVE ALL BURRS, SHARP EDGES AND SCALE REMOVED

(B) (7) WEDGE SHALL TOUCH CUSHION ON LOWER FOOT WITH CLAMP INSTALLED ON MANDREL WITHOUT SPACER

(B) (8) IF MIL-C-85052/4 CLAMP SUPPORTS ARE USED THE MAXIMUM TUBE OVERHANG SHALL BE 1/2 TUBE O.D THIS WASHER CAN BE DETRIMENTAL IF THE CLAMP IS NOT FULLY SUPPORTED.

(9) REFERENCE DIMENSIONS "Y" AND "J" ARE PROVIDED TO GAIN MAXIMUM SUPPORT FOR THE LOWER FOOT BY CLOSELY FITTING THE UPPER FOOT BEND RADIUS TO THE STIFFENING BEAD BLEND RADIUS WHEN CLOSED AGAINST EACH OTHER

REQUIREMENTS

MARKING

BAND - THE COMPLETE STANDARD PART NUMBER AND MANUFACTURER'S NAME, TRADEMARK, OR FSCM NUMBER SHALL BE IMPRESSION STAMPED ON THE BAND IN AN AREA NOT COVERED BY THE CUSHION ALL MARKING SHALL BE ABOVE TUBE THEORETICAL CENTERLINE

- CLAMP BANDS OF -2, -3, -4 SIZES MAY BE MARKED WITH THE MANUFACTURER'S IDENTIFICATION, THE SIZE, AND THE SPECIFICATION SHEET NUMBER DUE TO SPACE LIMITATIONS

CUSHION - NONE

MATERIALS

METAL BAND - 17-7PH, CORROSION RESISTANT STEEL PER MIL-S-25043, ANNEALED STRESS RELIEVE AND HEAT TREAT TO TH1100 CONDITION PER MIL-H-6875, AFTER FORMING

CUSHION - FABRIC REINFORCED, SILICONE RUBBER, 65-75 DUROMETER ON UNREINFORCED MATERIAL, COLOR LIGHT BLUE, PER THE GENERAL SPECIFICATION WITH ADDITIONAL REQUIREMENTS SHOWN ON THIS SPECIFICATION SHEET. THE FABRIC REINFORCING MATERIAL SHALL BE MOLDED INTO THE BAND SIDE OF THE CLAMP CUSHION AT A 45° ANGLE (SEE FIGURE A)

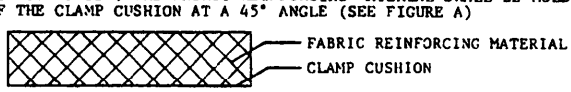


FIGURE A

FINISH

METAL BAND - PASSIVATE PER QQ-P-35

CUSHION - NONE

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PART NUMBER

PART NUMBERS SHALL CONSIST OF THE FOLLOWING (IN SEQUENCE)

1 THE LETTER "M"
2 THE GENERAL SPECIFICATION NUMBER
3 A SLASH, AND SLASH NUMBER OF THIS SPECIFICATION SHEET
4 A DASH, AND THE APPROPRIATE SIZE DASH NUMBERS FROM TABLE I

EXAMPLE M85052/3-8

CUSHION REQUIREMENTS

1 PHYSICAL PROPERTIES - AS SPECIFIED IN TABLE II UNLESS OTHERWISE SPECIFIED, RESULTS ARE AN AVERAGE OF 5 SPECIMENS AND TOLERANCE ON TEMPERATURE = ±5°F

TABLE II

TEST	TEST METHOD	REQUIRED ORIGINAL PROPERTIES	ALLOWABLE CHANGE FROM ACTUAL ORIGINAL PROPERTIES AFTER HEAT AGING 70 HRS AT +500°F	OIL IMMERSION IN MIL-L-7808 70 HRS AT +302°F	DIP TEST *
HARDNESS, DUROMETER "A"	ASTM D2240	65-75	+10 PTS MAX	-30 PTS MAX	+5 PTS
TENSILE STRENGTH - PSI	** ASTM D412	1200 MIN	1000 MIN.	300 MIN	800 MIN
ELONGATION - %	** ASTM D412	30 MIN	30 MIN	30 MIN	30 MIN
TEAR STRENGTH - PPI	ASTM D624 (DIE "B")	300 MIN	300 MIN	300 MIN	300 MIN
VOLUME CHANGE - %	ASTM D471	------	------	+45% MAX	+5% MAX
WEIGHT CHANGE - %	ASTM D471	REPORT	-5% MAX	------	------
SPECIFIC GRAVITY	ASTM D471	REPORT	------	------	------

(B) * DIP TEST - TEST SAMPLES SHALL BE DIPPED IN A-A-711, TYPE I SOLVENT AND ALLOWED TO DRY AT ROOM TEMPERATURE FOR 24 HRS. THIS PROCEDURE SHALL BE REPEATED 10 TIMES.

** TESTS SHALL BE CONDUCTED IN ACCORDANCE WITH ASTM D412 ON THE FABRIC REINFORCED CUSHION TEST SPECIMEN, EXCEPT AS NOTED HEREIN. DIE "A" DUMBELL SPECIMENS SHALL BE CUT FROM THE ACTUAL CUSHION WITH THE FABRIC WEAVE AT 45° TO THE DIRECTION OF PULL AND WITH THE EARS FOLDED OUT TO FLATTEN THE SPECIMEN. THE RATE OF SEPARATION OF THE JAWS SHALL BE 12 INCHES PER MINUTE. THE TENSILE STRENGTH AND PERCENT ELONGATION SHALL BE MEASURED WHEN THE FABRIC REINFORCING MATERIAL BEGINS TO SEPARATE FROM THE ELASTOMER OR THE SPECIMEN BREAKS COMPLETELY.

TENSILE STRENGTH SHALL BE BASED ON OVERALL THICKNESS OF RUBBER AND FABRIC REINFORCING MATERIAL. SPECIMEN FAILURES DUE TO FABRIC SEPERATION AT THE TEST FIXTURE JAWS SHALL BE DISCARDED AND ANOTHER SAMPLE TESTED.

(B) 2. COMPRESSION SET - SEE MIL-C-85052.
- AIR AGE AT 302°F.
- NOT TO EXCEED 30% AVERAGE OF 3 SPECIMENS.

(B) 3. FLAMMABILITY - SEE MIL-C-85052.
- SPECIMENS -16 SIZE CLAMP ASSEMBLIES UNBENT UNTIL FLAT.
- VERTICAL BURN TEST

INTENDED USE

THESE CLAMPS ARE INTENDED FOR USE AS FOLLOWS

TEMPERATURE RANGE

-65°F TO +500°F THESE CLAMPS HAVE CUSHIONS WHICH HAVE BEEN COMPOUNDED FOR HIGH TEMPERATURE CAPABILITY AT THE SACRIFICE OF FLUID RESISTANCE USE IS RECOMMENDED ONLY IN THOSE AREAS WITH OPERATING TEMPERATURES FROM 275°F TO 500°F WHERE FLUID EXPOSURE IS MINIMAL FOR APPLICATION BELOW 275°F THE USE OF CLAMPS WITH A HIGHER DEGREE OF FLUID RESISTANCE IS RECOMMENDED SUCH AS MIL-C-85052/1 OR /2 CLAMPS

SYSTEMS

ALL FLUID AND ELECTRICAL SYSTEM APPLICATIONS IN HIGH TEMPERATURE AREAS

VIBRATION RATING

DEPENDING ON MOUNTING METHODS (SEE RECOMMENDED MOUNTING DATA)

Custodians:
NAVY-AS
AIR FORCE-99
ARMY-AV

Reviewer activities
AIR FORCE - 11
DLA - IS

Preparing activity.
NAVY-AS

PROJECT NO 5340-1574

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RECOMMENDED MOUNTING METHODS

MOUNTING METHODS AFFECT THE VIBRATION CAPABILITIES OF THESE CLAMPS TYPICAL MOUNTING METHODS AND VIBRATION RATINGS ARE SHOWN IN FIGURES I, II, AND III THE METHOD SHOWN IN FIGURE II IS RECOMMENDED WHENEVER POSSIBLE THE METHOD SHOWN IN FIGURE III SHOULD ONLY BE USED IN VERY HIGH VIBRATION AREAS OR AS NECESSARY

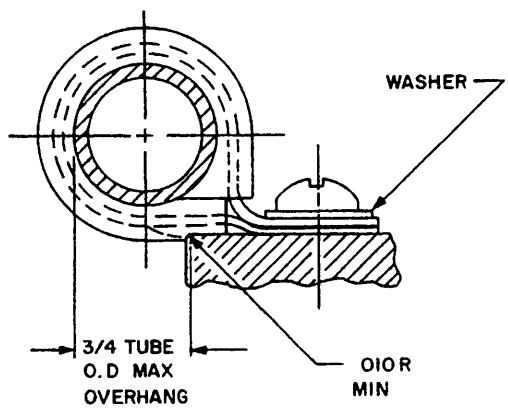


FIGURE I
LOW TO MEDIUM VIBRATION AREAS

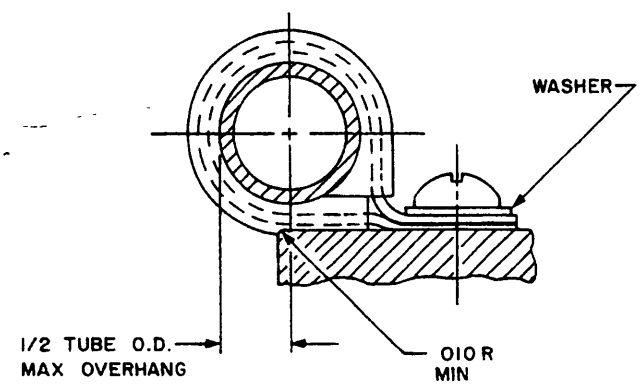


FIGURE II
LOW TO HIGH VIBRATION AREAS

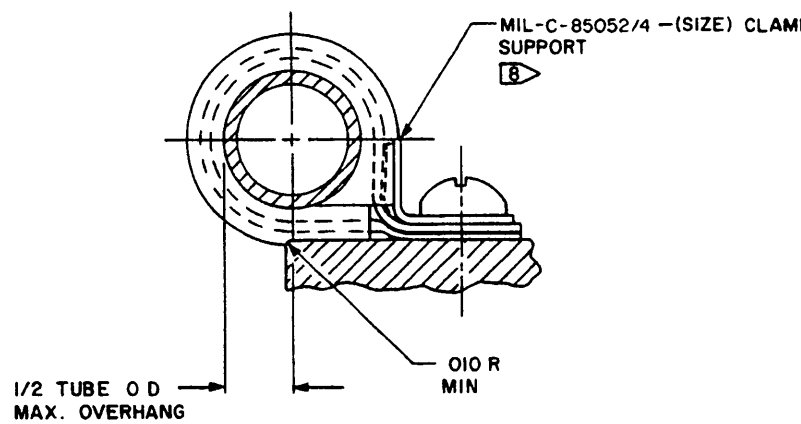


FIGURE III
VERY HIGH VIBRATION AREAS

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INCH–POUND

NOTICE
OF VALIDATION

MIL–C–85052/3B
NOTICE 1
26 September 1991

MILITARY SPECIFICATION

CLAMP, LOOP, TUBE 17–7 PH CRES, 500 DEG. F.

MIL–C–85052/3B, dated 30 May 1984, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army – AV
Navy – AS
Air Force – 99

Preparing activity.
Navy – AS

AMSC N/A FSC 5340

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